PRE-ALGEBRA (7TH) · UNIT 13: NONLINEAR FUNCTIONS AND POLYNOMIALS

Rabbit Island Population

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students manage an island rabbit population. Survey data introduces geometric sequences and exponential growth, foxes introduce decay, and fenced habitats require polynomial operations.

Comparing linear grass loss with exponential rabbit growth shows why nonlinear models matter.

Standards Addressed

- 8.F.A.3 Linear vs. nonlinear functions
- F-LE.A.1–2 Linear and exponential models (preview)
- A-APR.A.1 Add, subtract, and multiply polynomials
- F-BF.A.2 Arithmetic and geometric sequences

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a news story about invasive rabbits. Part 1 sequences and patterns.
Day 2	Part 2 graphs and exponential models.
Day 3	Part 3 habitat polynomials and the report.

Materials

- Student handout and Population Survey
- Graph paper
- Calculators
- Optional: graphing tool

How the Three Versions Differ

Level	What changes
CP2	Doubling patterns, arithmetic vs. geometric, graphs, one exponential rule, and adding perimeters.
CP1	Growth and decay models, comparing linear, quadratic, and exponential, and multiplying binomials.
Honors	Fitting the model, carrying capacity, solving exponential inequalities by testing, sequences vs. demand, and optimizing area.

Watch For (Common Misconceptions)

- Thinking $10 \cdot 2^n$ means 20^n.
- Writing $(x + 5)^2$ as $x^2 + 25$.
- Using 0.3 instead of 0.7 for a 30% decrease.

Extension Ideas

- Research the real rabbits of Australia.
- Model a different animal with a growth rate.
- Use a spreadsheet to graph 20 years.

Answer Key – CP2

Part 1: Patterns

#	Task	Answer
1	What do you multiply by each year?	2
2	Predict the population in years 5 and 6.	320 and 640
3	Grass patches drop by 15 each year: 200, 185, 170, … Is this arithmetic or geometric? What comes next?	Arithmetic; 155
4	In which year will the grass patches reach 50?	200 − 15n = 50 → n = 10, year 10

Part 2: Graph It

#	Task	Answer
5	Graph the population for years 0–4. Is the graph a line or a curve?	A curve that gets steeper (exponential).
6	Graph the grass patches for years 0–4. Is it a line or a curve?	A line going down (linear).
7	Write a rule for the population after n years.	$P = 10 \times 2^n$
8	Use your rule to find the population in year 8.	$10 \times 2^8 = 2{,}560$

Part 3: Habitat Fences

#	Task	Answer
9	Write and simplify the perimeter of the pen.	$2(x + 4) + 2x = 4x + 8$
10	Add the perimeters of two pens: $(4x + 8) + (2x + 6)$.	$6x + 14$
11	If $x = 5$, how much fence is needed for both pens?	$6(5) + 14 = 44$ m

Reflection

#	Prompt	Look for
R1	How is doubling different from adding the same number each year?	Doubling grows faster and faster; adding grows by the same amount.
R2	Why can the rabbit population not keep doubling forever?	The island runs out of food and space.

Answer Key – CP1

Part 1: Growth Models

#	Task	Answer
1	Write the population as a function P(n) and find P(7).	$P(n) = 10(2)^n$; 1,280
2	Each rabbit eats 0.1 grass patch per year. Make a table for years 4–8 comparing the grass patches (200 − 15n) with the grass the rabbits need (0.1 × P(n)). In which year do the rabbits first need more grass than the island has?	Year 7: the rabbits need 128 patches but only 95 remain (year 6: 64 needed, 110 available).
3	Biologists bring foxes, and the rabbits now decrease by 30% per year from 640. Write a decay model and find the population after 3 years.	$P = 640(0.7)^n$; ≈ 219.5 → about 220 rabbits
4	Is the decay model linear? Explain using the differences in your table.	No; it decreases by 30% of a shrinking amount, so differences are not constant.

Part 2: Compare Functions

#	Task	Answer
5	Graph $y = 10 \cdot 2^x$ and $y = 50x + 10$ for x = 0 to 5. Where does the exponential pass the linear?	Between x = 4 and x = 5 (x = 4: 160 vs 210; x = 5: 320 vs 260).
6	Graph $y = x^2$ for the area (sq m) of a square pen with side x. What shape is the graph?	Parabola (U-shaped) — first quadrant half
7	Which grows fastest eventually: 50x + 10, x^2, or $10 \cdot 2^x$?	$10 \cdot 2^x$
8	Write the type of each sequence: 3, 9, 27, …; 5, 12, 19, …; 1, 4, 9, 16, …	Geometric (×3); arithmetic (+7); neither (squares)

Part 3: Habitat Design

#	Task	Answer
9	Write and simplify the area as a polynomial.	$(2x + 3)(x + 1) = 2x^2 + 5x + 3$
10	A larger pen is (3x + 2)(x + 4). Find the difference in area between the larger pen and the first pen.	$3x^2 + 14x + 8 - (2x^2 + 5x + 3) = x^2 + 9x + 5$
11	If x = 6, find both areas and the total fence for both pens.	Areas 105 and 200 sq m; perimeters 2(15 + 7) = 44 and 2(20 + 10) = 60 → 104 m

Reflection

#	Prompt	Look for
R1	How can you tell from a table whether a function is exponential?	Consecutive outputs have a constant ratio.
R2	How is multiplying binomials like finding the area of a rectangle?	Each part of one length multiplies each part of the other, like splitting the rectangle into smaller rectangles.

Answer Key – Honors

Part 1: Model the Data

#	Task	Answer
1	Show that the survey data is exponential by finding ratios. Write P(n).	Each ratio is 2; $P(n) = 10 \cdot 2^n$
2	In what year does the population first exceed 10,000?	$2^n > 1{,}000 \rightarrow n = 10$ (10,240)
3	The island can support at most 3,000 rabbits. Explain why the model must change before year 9. What might the real graph look like?	$P(9) = 5{,}120 > 3{,}000$; growth slows and levels off (an S-shaped curve).
4	After foxes arrive, the population falls 30% per year from 2,500. After how many whole years is it below 500?	$2{,}500(0.7)^n < 500 \rightarrow (0.7)^n < 0.2 \rightarrow n = 5$ (≈ 420)

Part 2: Sequences & Graphs

#	Task	Answer
5	Write the nth term of the grass sequence 200, 185, 170, … and of the rabbit sequence 10, 20, 40, …	Grass $a_n = 200 - 15n$ (n starting at 0); rabbits $10 \cdot 2^n$
6	Each rabbit eats 0.1 grass patch per year. In what year does the rabbits' need exceed the grass patches?	$0.1 \cdot 10 \cdot 2^n > 200 - 15n$: n = 7 → 128 > 95 ✓; n = 6 → 64 < 110. Year 7.
7	Sketch $y = x^2$, $y = x^3$, and $y = 2^x$ for $0 \le x \le 4$ on one graph. Where does 2^x pass x^2?	They meet at x = 2 and x = 4; for $x > 4$, $2^x > x^2$.
8	Explain why an exponential function eventually outgrows any polynomial.	It multiplies by a constant factor each step, while polynomial growth rates increase more slowly.

Part 3: Habitat Optimization

#	Task	Answer
9	A pen has length $3x + 2$ and width $2x - 1$. Write its area and perimeter as polynomials.	Area $6x^2 + x - 2$; perimeter $10x + 2$
10	Square the binomial: a square pen has side $x + 5$. Find its area and explain why it is not $x^2 + 25$.	$x^2 + 10x + 25$; the two middle rectangles (5x each) are missing from $x^2 + 25$
11	You have 104 m of fence for one rectangular pen. Write the area of a pen with width w as a polynomial and find the width that gives the most area (test values).	Length $52 - w$; area $52w - w^2$; maximum at w = 26 (676 sq m)

Reflection

#	Prompt	Look for
R1	Why are real populations not exponential forever?	Limited food and space slow growth; models apply only over certain domains.
R2	What pattern did you notice when maximizing the pen area?	A square (equal length and width) gives the greatest area for a fixed perimeter.